The Human Auditory System

SYNOPSIS

Sound arrives at you as a pressure variation in air, and what reaches your brain is a nerve signal. Something has to do that conversion, and it is the ear. This note follows one sound through it, from the outside of your head to the nerve, and asks what each part is for.

PERFORMANCE EXPECTATIONS

Identify the major components of the auditory system and say which of the three divisions each one belongs to. Describe what each division does to the sound: the outer ear collecting it, the middle ear passing it from air into fluid, the inner ear turning it into nerve impulses. Explain why a boundary between air and fluid needs matching at all, and what the middle ear does about it. Describe how loud sound causes hearing loss, and ways to protect your hearing.

Instructions and Questions

Module 2 ended with sound as a pressure wave travelling through air. This note is about what happens when that wave runs into the side of your head. Nothing in here changes the sound — the ear does not make it louder or alter its frequency. The ear's whole job is **conversion**: pressure in air at one end, electrical impulses on a nerve at the other.

Three parts, one chain

The ear is usually divided into three, and the divisions are not arbitrary — each one works in a different medium. The **outer ear** is open to the air. The **middle ear** is an air-filled cavity behind the eardrum. The **inner ear** is filled with fluid. Sound has to cross all three.

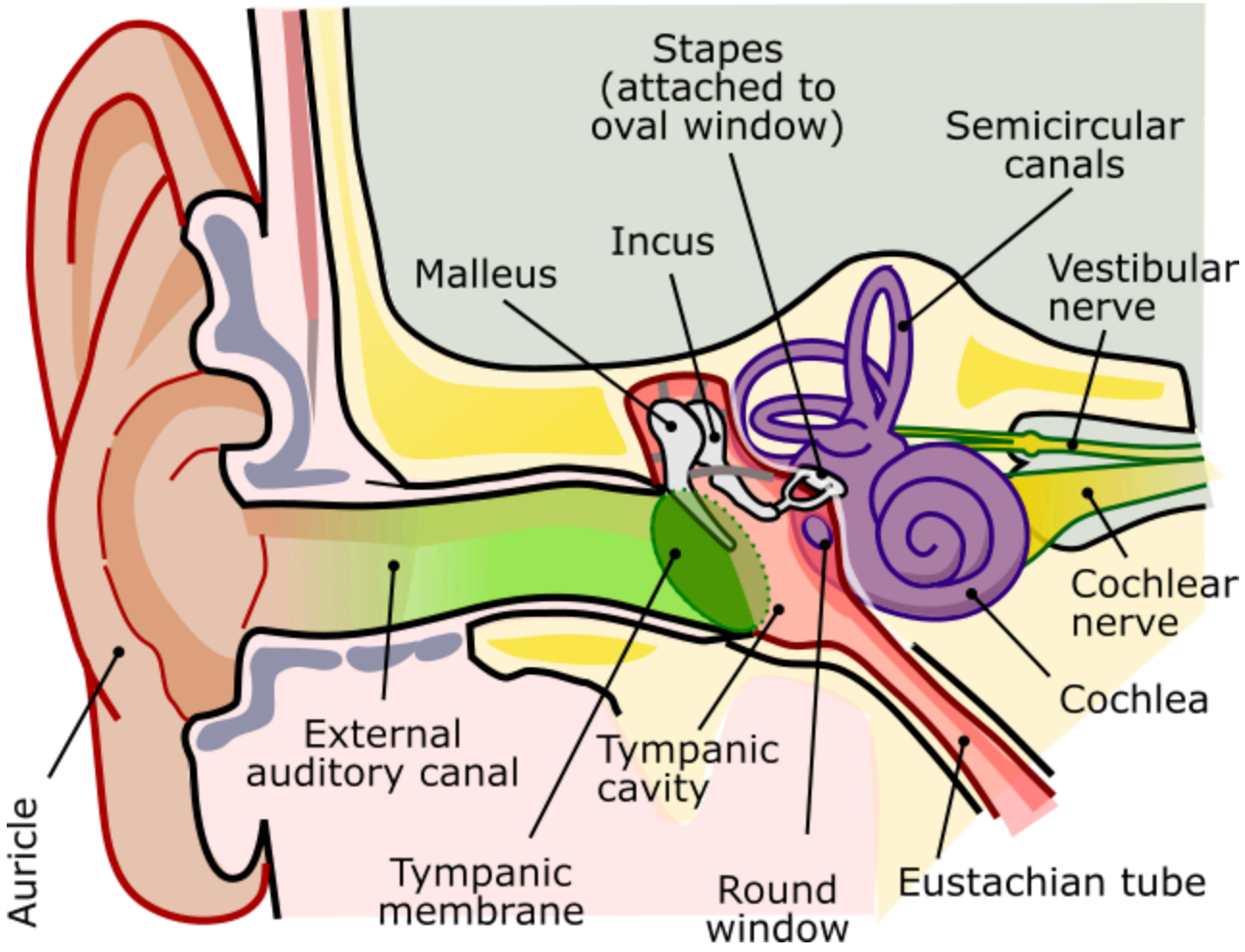


Figure 1. A section through the ear. Follow it left to right and you are following one sound: in at the auricle, down the canal, into the eardrum, across the three small bones, through the oval window into the fluid of the cochlea, and out along the nerve. The three divisions are the three media it crosses.

Figure after Chittka L, Brockmann, *Anatomy of the Human Ear*, CC BY 2.5, via Wikimedia Commons; numbered version derived by Sgbeer. Same figure used in the PHY 105 reference deck.

1 What are the three major parts of the ear, and what is the one thing that makes them three different parts rather than one?

> **Outer, middle and inner.** What separates them is the **medium the sound is travelling in**: air in the outer ear, air in the middle-ear cavity but carried by bone rather than by the air itself, and **fluid** in the inner ear. The boundaries between them are the two membranes — the tympanic membrane and the oval window.

The outer ear: collecting

The **auricle** is the fleshy part you can see. The **ear canal** — the external auditory canal — is the tube running from the outside world in to the **tympanic membrane**, which is the eardrum. In your own words from last year: the outer ear *collects sound and funnels it into the inner parts*. That is the whole of its job.

2 You cup a hand behind your ear to hear someone better across a room, and it works. What have you actually changed? Be careful: you have not made the talker any louder.

> You have made the **collecting area bigger**. The sound energy arriving is spread over a certain area; a bigger funnel gathers more of it and sends it down the same canal, so more energy arrives at the same eardrum. The talker is unchanged — you have changed how much of what they sent you is caught. A cupped hand also blocks sound from behind you, which helps as much as the gathering does.

LOOKING AFTER THE CANAL

The canal is a dead end sealed by a membrane you cannot replace, and nothing rigid should go down it. If it needs cleaning, that is drops and a rubber bulb — not a cotton bud, which packs wax against the eardrum, and not anything you could light on fire.

The middle ear: passing it on

Behind the eardrum is the **tympanic cavity**, and bridging it are the three smallest bones in your body: the **hammer, anvil and stirrup** — the malleus, incus and stapes. They are hinged to each other so that they work as a **lever**. One end rests on the eardrum; the far end, the footplate of the stirrup, rests on the **oval window**, which is the door into the inner ear.

So the conversion in this division is **sound wave → mechanical vibration**. Air pushes the eardrum, the eardrum pushes the bones, the bones push the oval window.

One more opening matters here. The **Eustachian tube** runs from the tympanic cavity down to the back of the throat, and it exists to **equalise the pressure** on the two sides of the eardrum.

3 An aeroplane climbs and your ears block up. Swallowing or yawning fixes it. Using the Eustachian tube, say what went wrong and what swallowing did. Then say why your hearing was worse while it lasted.

> Cabin pressure fell as you climbed, but the air sealed in the tympanic cavity did not, so there was **more pressure inside than outside** and the eardrum was pushed outward and held taut. Swallowing opens the Eustachian tube for a moment and lets the trapped air out to the throat, so the two sides match again. Hearing is worse while it lasts because a membrane under tension cannot vibrate freely — it is stiff, so less of the sound gets through.

Why the middle ear has to be there at all

It is worth asking why the eardrum does not simply sit on the fluid. Why is there a cavity with a lever in it at all? The answer is the one idea in this note that is pure physics, and it is called **impedance**.

A medium's **acoustic impedance** is how hard it is to get it moving — roughly, its density times the speed of sound in it. Air is light and easy to move. Water is about 3500 times harder. When a sound wave meets a boundary between two media whose impedances are very different, almost all of it **reflects** rather than crossing. At a plain air-to-water boundary, around **99.9%** of the energy bounces straight back.

The inner ear is full of fluid. So a sound going straight from air into it would almost entirely bounce off, and you would be very nearly deaf. The middle ear is the fix. It is an **impedance matcher**, and it works two ways at once.

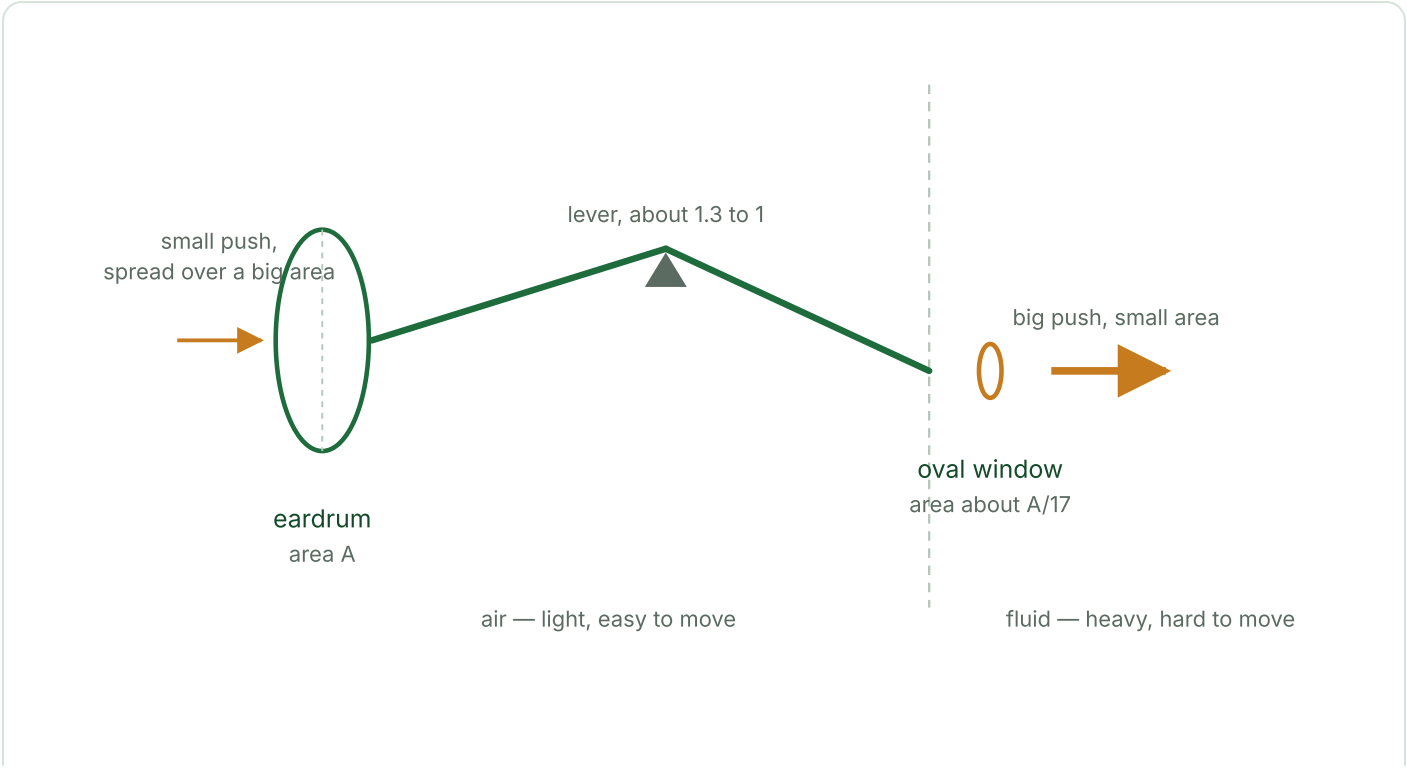


Figure 2. Drawn to the real ratios. The eardrum has roughly **17 times the area** of the oval window, so the same total force is delivered onto a seventeenth of the area — and pressure is force over area. The ossicles add a **lever** of about 1.3 to 1 on top of that. Together they raise the pressure at the oval window enough to drive the fluid, which is exactly what a light, easy medium cannot otherwise do to a heavy, hard one.

4 The eardrum has about 17 times the area of the oval window. Take the same total force arriving at each. What does that do to the **pressure**, and why is pressure the thing that matters here rather than force?

> Pressure is force divided by area, so the same force on a seventeenth of the area is about **17 times the pressure**. Pressure is what matters because it is what pushes on the fluid: the fluid does not care how much total force the eardrum collected, only how hard it is being pushed at the window where it is being pushed. Concentrating a modest force onto a small window is how a weak push in air becomes a push strong enough to move fluid.

5 Someone's ossicles have fused and can no longer move. Sound still reaches the eardrum and the inner ear is perfectly healthy. Can they hear? Explain using the figure.

> Barely, and this is **conductive** hearing loss. The matcher is gone, so the sound meets the air-to-fluid boundary with nothing to concentrate it, and almost all of it reflects. The hair cells are fine and the nerve is fine — the signal simply never gets in to them. It is the same reason you hear so little of what is said above the surface when your head is under water.

The inner ear: making the signal

The **cochlea** is a fluid-filled, snail-shaped spiral — your own description of it. The stirrup drives the **oval window** at one end; the **round window**, just below it, bulges out as the oval window pushes in. Running the length of the spiral is the **basilar membrane**, and sitting on that is the **organ of Corti**.

The organ of Corti carries **hair cells**, named for the **stereocilia** standing up from them. When the fluid moves, the membrane above them shears across those stereocilia and bends them, and a bent hair cell fires. That is the moment the signal stops being mechanical: **mechanical vibration → electrical nerve impulse**, out along the cochlear nerve to the brain.

6 The fluid in the cochlea cannot be squashed. Given that, say why there has to be a round window, and what would happen without one.

> Fluid is effectively **incompressible**, so if the stirrup pushes the oval window inward, that fluid has to go somewhere. The round window is the somewhere — it bulges outward by the same amount at the same instant. Without it the fluid would have nowhere to go, the oval window could not move at all, and nothing would travel down the cochlea. A second flexible window is not a detail; it is what makes the first one able to move.

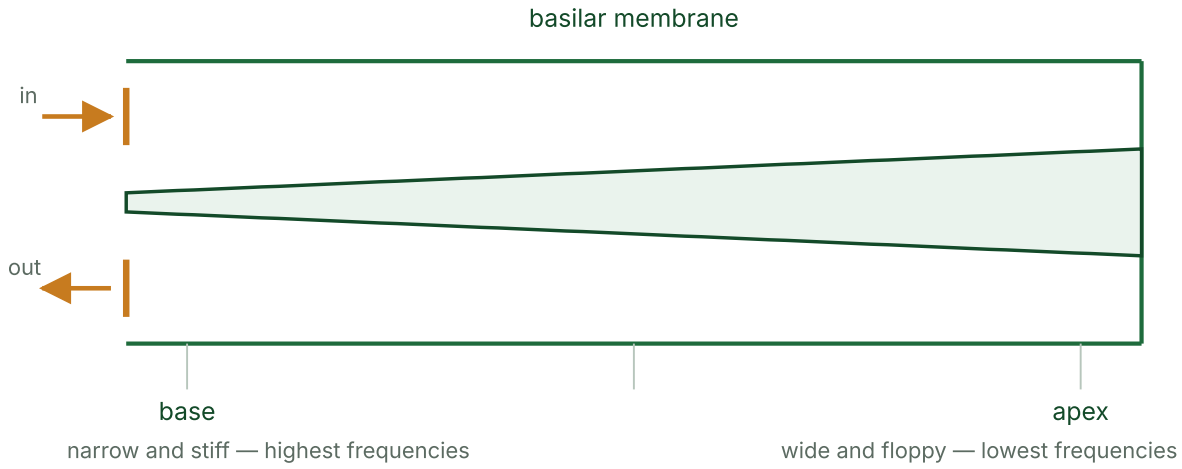


Figure 3. The cochlea drawn uncoiled. The basilar membrane is **narrow and stiff at the base**, where the windows are, and **wide and floppy at the apex**. A stiff, narrow strip responds best to fast vibration and a wide, floppy one to slow vibration, so each frequency sets a different stretch of the membrane moving most. **Position along the cochlea stands for frequency**, and which hair cells fire is what tells the brain the pitch.

7 Where along the membrane does a high-pitched whistle do most of its work — near the base or near the apex? And what does the brain actually use to tell one pitch from another?

> Near the **base**, where the membrane is narrow and stiff. Low notes travel further in and peak near the **apex**. What the brain uses is **which hair cells sent the signal** — not anything about the signals themselves, which all look alike. Pitch is encoded as **place**.

8 The semicircular canals sit right next to the cochlea, are filled with the same kind of fluid, and send signals along a nerve of their own. They have nothing to do with hearing. What are they for, and why is it reasonable that they ended up here?

> **Balance.** They are three fluid-filled loops set at angles to each other, and when your head turns, the fluid lags behind and bends hair cells of the same kind, which report the movement to the brain along the **vestibular nerve**. It is reasonable that they are here because they are the same machine: a fluid that moves, and hair cells that report that movement. Hearing and balance are two uses of one invention.

Hearing loss, and keeping yours

Everything above the hair cells is replaceable in principle — eardrums can be patched, ossicles can be replaced with prosthetic ones. The hair cells themselves are the problem: **you are born with all of them you will ever have, and they do not grow back.**

Loud sound damages them mechanically — the stereocilia are bent too far, too often, and break. Two things decide the damage: **how loud** and **for how long**. A short, very loud event and a long, moderately loud one can do the same harm, which is why a day at work in a noisy room is as real a risk as a concert. The loss shows up first in the **high** frequencies, because the base of the cochlea takes the brunt of every sound going in.

9 Noise-induced hearing loss usually appears first at high frequencies, around 4000 Hz, whatever the noise was. Using Figure 3, suggest why the base of the cochlea should suffer first.

> Every sound entering the cochlea passes the **base** on its way in, whatever its frequency, so the hair cells there are worked by all of it and take the most punishment over a lifetime. Only low frequencies reach the apex strongly. So damage accumulates where the traffic is, and the high-frequency cells are the ones standing in it.

10 Name three different ways to cut the dose your hair cells take. You should be able to get all three straight out of the two things that decide the damage.

The dose is level × time, so: **turn it down** (level), **spend less time in it** and take quiet breaks (time), and **put distance or a barrier in the way** — step back from the speaker, or wear plugs or muffs, which lowers the level that reaches you. Good plugs lower everything roughly evenly rather than muffling, so musicians can still hear what they are playing.

11 A cochlear implant does not make sound louder. It runs a wire of electrodes into the cochlea and stimulates the nerve directly. From the chain in Figure 1, say which parts it is replacing and which part it absolutely depends on still working.

It replaces **everything up to and including the hair cells** — outer ear, eardrum, ossicles, oval window and the conversion step itself. What it depends on is the **cochlear nerve**: it is feeding that nerve directly, so there has to be a working nerve to feed. And because place stands for frequency, the electrodes are spread along the cochlea so that each one stimulates the spot where its own band of frequencies belongs.

WHERE THIS IS GOING

Every question in this note has been about *how* the ear works, not *how much* it takes to hear something. Note 2 puts a number on that. Your ear covers a range of intensities so enormous that ordinary numbers become useless for it, which is where the decibel comes from.

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